



LAB ASSIGNMENT

SUBMITTED TO:

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Course: Higher Level Programming Language Lab

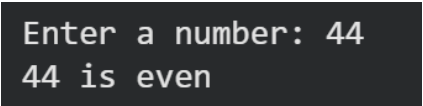
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If-else statement

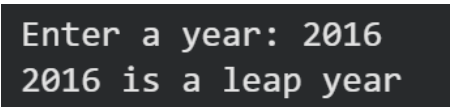
Problem 1: Write a program that takes an integer input and prints whether it is even or odd.

```
num = int(input('Enter a number: '))  
if num%2==0:  
    print(f'{num} is even')  
else:  
    print(f'{num} is odd')
```

Output: 

Problem 2: Write a program to determine whether a given year is a leap year.

```
num = int(input('Enter a year: '))  
if (num%4==0 and num%100!=0) or (num%100==0 and num%400==0):  
    print(f'{num} is a leap year')  
else:  
    print(f'{num} is not a leap year')
```

Output: 

Problem 3: Given three angles, determine if they can form a valid triangle.

```
ang1 = int(input('Enter angle 1: '))  
ang2 = int(input('Enter angle 2: '))  
ang3 = int(input('Enter angle 3: '))  
  
if (ang1>0 and ang2>0 and ang3>0) and (ang1+ang2+ang3==180):  
    print('It is valid')
```

else:

print('It is not valid')

Output:

```
Enter angle 1: 30
Enter angle 2: 60
Enter angle 3: 90
It is valid
```

Problem 4: Check whether a number is: Positive, Negative or Zero

num = int(input('Enter a number: '))

if num>0:

print(f'{num} is positive')

elif num==0:

print(f'{num} is zero')

else:

print(f'{num} is negative')

Output:

```
Enter a number: 55
55 is positive
```

Problem 5: Perform arithmetic operations on two numbers.

num1 = int(input('Enter first number: '))

num2 = int(input('Enter second number: '))

op = input('Enter operator: (+,-,*,/)')

if op=='+':

result = num1+num2

elif op=='-':

result = num1-num2

elif op=='*':

result = num1*num2

```
else:
if num2==0:
print('Invalid operation')
exit()
else:
result = num1/num2
print(result)
```

Output:

```
Enter first number: 56
Enter second number: 55
Enter operator: (+,-,*,/)-
1
```

Looping problems

Problem 1: Sum of First N Natural Numbers. Input N and calculate: $1+2+3+...+n$

```
num = int(input('Enter the last number of the series: '))
sum=0
for i in range(1,num+1):
sum = sum+i
print(f'The sum of {num} numbers: {sum}')
```

Output:

```
Enter the last number of the series: 10
The sum of 10 numbers: 55
```

Problem 2: Calculate factorial using loop.

```
fact = int(input('Enter the number: '))
res = 1
for i in range(1,fact+1):
res = res * i
print(f'The factorial of {fact}: {res}')
```

Output: Enter the number: 7
The factorial of 7: 5040

problem 3: Check whether a string is a palindrome.

```
str = input('Enter the string: ')
flag = True
for i in range(len(str) // 2):
    if str[i] != str[len(str)-i-1]:
        print('Not a palindrome')
        flag=False
        break
if flag==True:
    print('It is a palindrome')
```

Output: Enter the string: hannah
It is a palindrome

Problem 4: Find the Fibonacci Series of a given number

```
num = int(input("Enter the number of elements: "))
frst = 0
sec = 1
print(frst, sec, end=' ')
for i in range(2, num):
    temp = frst + sec
    print(temp, end=' ')
    frst = sec
    sec = temp
```

Output: Enter the number of elements: 10
0 1 1 2 3 5 8 13 21 34

Problem 5: Count even and odd numbers in a range

```
start = int(input("Enter the start of the range: "))
end = int(input("Enter the end of the range: "))

ec=0
oc=0

for num in range(start, end + 1):
    if num % 2 == 0:
        ec += 1
    else:
        oc += 1

print(f"Even numbers count: {ec}")
print(f"Odd numbers count: {oc}")
```

Output: Enter the start of the range: 10
Enter the end of the range: 100
Even numbers count: 46
Odd numbers count: 45

Function definition

Problem 1: Write a problem to determine if a number is prime or not

```
def prime_check(n):
    if n <= 1:
        return False

    for i in range(2, int(n**0.5) + 1):
        if n % i == 0:
            return False
```

```
return True
```

```
num = int(input('Enter a number: '))
```

```
ans = prime_check(num)
```

```
if ans==True:
```

```
print('It is a prime number')
```

```
else:
```

```
print('It is not a prime number')
```

Output:

```
Enter a number: 67
It is a prime number
```

Problem 2: Write a function to reverse a string without using reverse functions

```
def reverse_string(s):
```

```
    reversed_s = ""
```

```
    for char in s:
```

```
        reversed_s = char + reversed_s
```

```
    return reversed_s
```

```
str = input('Enter the string: ')
```

```
print(f'The reversed string: {reverse_string(str)}')
```

Output:

```
Enter the string: icee
The reversed string: eeci
```

Problem 3: Write a function to find the largest number in a list

```
def find_max(lst):
```

```
    if not lst:
```

```
        return None
```

```
maximum = lst[0]

for num in lst:
    if num > maximum:
        maximum = num

return maximum

l = [5,2,6,1,8]

print(f'The largest number in the list is: {find_max(l)}')
```

Output: The largest number in the list is: 8

Problem 4: Find the factorial of a number

```
def factorial(n):
    if n < 0:
        return None
    if n == 0 or n == 1:
        return 1
    return n * factorial(n - 1)

num = int(input('Enter the number: '))

print(f'The factorial of {num} is: {factorial(num)}')
```

Output: Enter the number: 7
The factorial of 7 is: 5040

Problem 5: Write a function to determine the number of vowels in a string.

```
def count_vowels(s):
```

```
    vowels = "aeiouAEIOU"
```

```
    count = 0
```

```
    for char in s:
```

```
        if char in vowels:
```

```
            count += 1
```

```
    return count
```

```
str = input('Enter a string: ')
```

```
print(f'The number of vowels in the string: {count_vowels(str)}')
```

Output:

```
Enter a string: Maisha Binte Monir
The number of vowels in the string: 7
```